Micro-Python Final Project Tutorial

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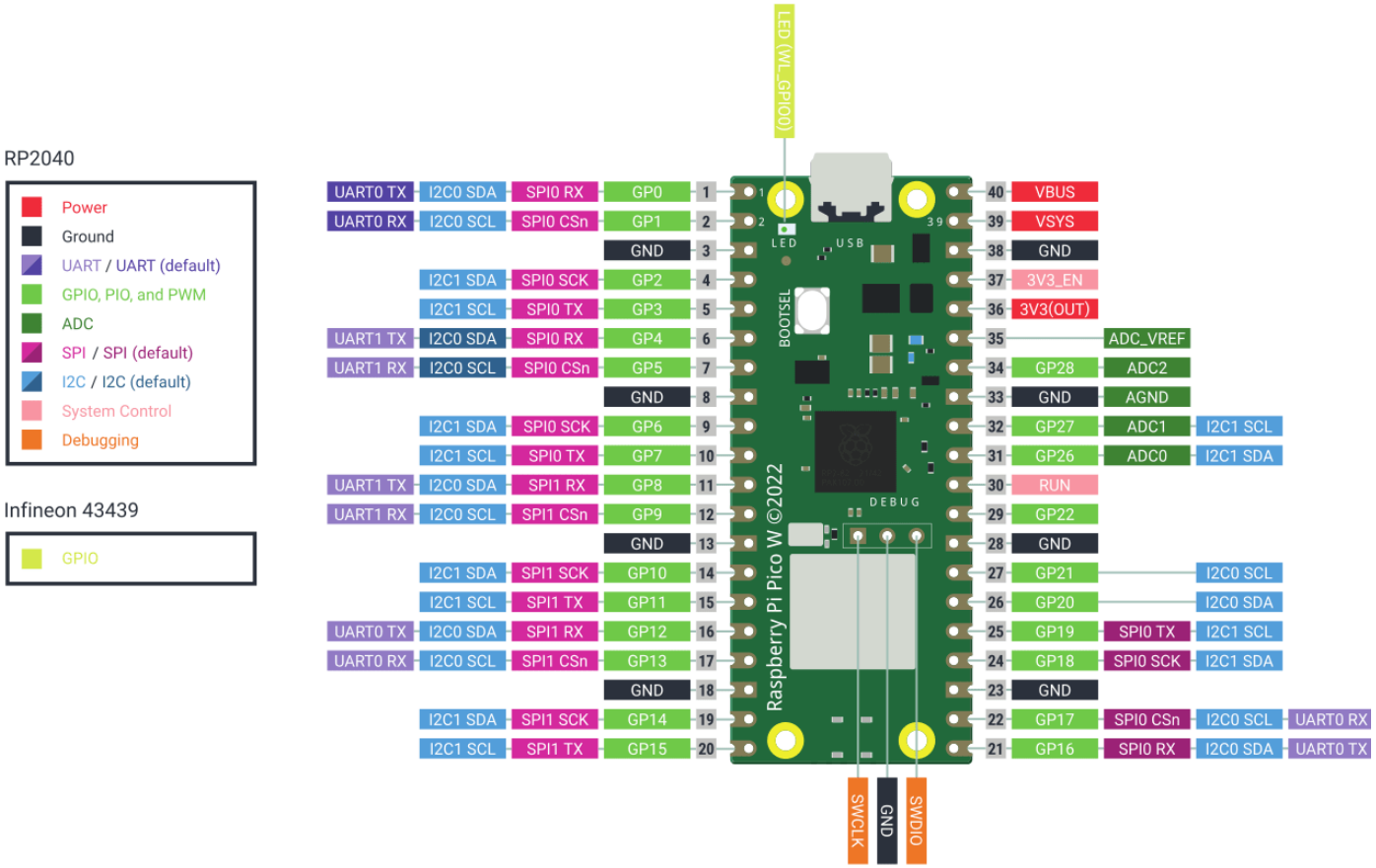
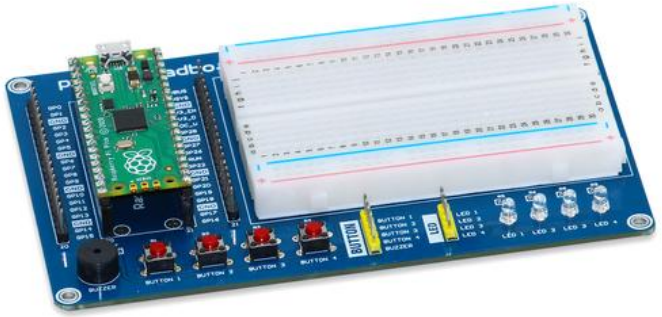
Final Project – Utilizing a Raspberry Pi Pico (Micro-Python)





Possible Themes for ECE 1000 Final Project

Automatic Plant Watering System:

- **Components:**
  - Raspberry Pi Pico
  - Soil Moisture Probe
  - Water Tank
  - Water Pump, Relay, & Tubes

- **Goals (Continually – as a while loop):**
  - Measure the water & moisture level of the soil.
  - Tell the Raspberry Pi Pico to send water if needed.
  - If not needed, continue measuring the water level.
  - **Override** and send water to the tank upon user input.
  - You must have a **clean** system (pot, 3D printed, etc).

Final Project – 3D Printed Robot Arm

Joystick Controlled Robot Arm (Servo Motors)

- **Components:**
  - Raspberry Pi Pico
  - Servo Motors
  - Joystick
  - 3D Printed Chassis

- **Goals (Continually – as a while loop):**
  - Create a low-cost robot using a 3D printed chassis and C.O.T.S. motors.
  - Use 3 or 4 servos to control the rotation, vertical, and claw movement.
  - Use a joystick to control the servo motors of the robot arm.
  - Must be able to turn **left to right**, **move up & down**, and **open & close a claw**.
  - The chassis must be 3D printed (via Library Maker Space or your own printer).
  - You can use Open-Source 3D printing websites such as Thingiverse/Grab CAD

Raspberry Pi Pico Microcontroller and Breakout Boards

- The Raspberry Pi (R-Pi) Pico is a **microcontroller** (like an **Arduino**) that allows users to implement logic to control different actuators and Input/Output (I/O) peripherals such as:
  - Motors
  - Switches
  - LEDs
  - Sensors
  - etc…
- We use the Micro-Python programming language (a variation of Python) to program R-Pi Picos.
- The Breakout Board allows users to easily connect different sensors and devices to the R-Pi Pico.

Signing Out a Raspberry Pi Pico Kit – Final Project

ECE 1000

Explorations in ECE

Fall 2025

DEVICE NUMBER:

Device required for course: Raspberry Pi Kits

By signing below, I (group leader) acknowledge receipt of the above noted device(s). I understand that all devices and associated peripherals are to be returned to the ECE 1000 instructors (or TA) by the last day of classes, Friday, December 12, 2025. I understand that failure to return this equipment in a usable condition with all peripherals by the due date may result in consequences including:

- Charging my student account for the full replacement cost of the device(s).
- Holding my grades in the above course(s).
- Disciplinary action.

Signature: _____ Date: _____

Printed Name: _____

T Number: _____

TTU Email: _____

Teammate 2 Name: _____

Teammate 3 Name: _____

Teammate 4 Name: _____

FOR OFFICE USE ONLY:

* ID/enrollment verified, and equipment issued: _____

* Returned device in satisfactory condition: _____

Getting Started – Raspberry Pi (R-Pi) Pico WH Kit

- First, open your Raspberry Pi Pico Kit and remove the board from anti-static wrapper.
- Next, plug the Micro-USB side of the cable into the Raspberry Pi (RPi) Pico.
- Next, if you don't have the "Thonny" IDE installed, see the following link and install it:
  - **Windows - Install with 64-bit Python 3.10**
  - **Mac - Install with Python 3.10**
  - **Thonny –** https://thonny.org/
- Next, hold the "BOOTSEL" button on the RPi Pico and plug the Micro-USB into your computer.
  - **The RPi Pico won't register correctly unless you hold down the "BOOTSEL" button!**

Programming the Raspberry Pi Pico WH – Blinking LED

blinking_led.py

```python
import machine
import utime

led = machine.Pin("LED", machine.Pin.OUT)

while True:
    led.off()
    utime.sleep(2)
    led.on()
    utime.sleep(2)
```

Incorporating the Breakout Board With the Raspberry Pi Pico

main.py

```python
from PicoBreadboard import LED,BUZZER,BUTTON
import time

#create object for various class
LED1 = LED(0) #create object for LED class, LED connected at GP0
BT1 = BUTTON(4) # Button connected at GP4
buzzer = BUZZER(8) # Buzzer connected at GP8

while 1:
   if BT1.read() == 1: #button pressed
       LED1.on()   #led 1 on
       buzzer.on() # buzzer on
       print("LED 1 ON")
   else :
       LED1.off()
       buzzer.off()
       print("Press Button 1")

   time.sleep(0.5) # delay 500ms
```

Final Remarks

- Have Fun! We don't expect you to be a programming master, however, we do expect you to have the ability to conduct research on your own (textbooks, YouTube, etc). We are also here to help!
- If you use anything someone else created, you must cite it! This also includes ChatGPT and other coding resources!
- Remember, the project is worth 20% of your final grade! Make sure to start now and work hard!
- All project teams must complete the assignment in Ilearn declaring their project TODAY!
- If your team does not return ALL materials after the demo, you will receive 0% for the project grade!
- The following is the submission link for the TTU Makerspace 3D Printing: https://ttuicube.com/3dprint

Thank You!